Pearson Edexcel Level 1/Level 2 – style practice paper

GCSE (9–1) Mathematics

1MA1/1H & 2H – Higher Tier

Topic 6: Statistics

Centre Number		**Candidate Number**	
Surname		**Other names**	

Practice questions in the style of Pearson Edexcel GCSE Mathematics (1MA1). Every question is original; the layout, command words, mark allocations and mark scheme (M, A, B marks) follow the real exam format. Calculator rules are given in the instructions below.

Time allowed	About 85 minutes (roughly 1 minute per mark)
Instructions	• Use **black** ink or ball-point pen. • Answer **all** questions. • Answer the questions in the spaces provided – there may be more space than you need. • Calculators must **not** be used in Section A. Calculators may be used in Section B. • Diagrams are **NOT** accurately drawn, unless otherwise indicated. • You must **show all your working**. • **Unless stated, give non-exact answers to 3 significant figures.**
Information	• The total mark for this paper is **85**. • The marks for **each** question are shown in brackets – use this as a guide as to how much time to spend on each question. • Questions labelled with an asterisk (*) are ones where the quality of your written communication will be assessed.
Advice	• Read each question carefully before you start to answer it. • Keep an eye on the time. • Try to answer every question. • Check your answers if you have time at the end.

Name: _____ Date: _____ Mark: _____ / 85

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Section A: Non-calculator

You must not use a calculator for questions 1 to 13.

1 Here are eight numbers. 3 7 4 9 7 5 7 10

(a) Work out the mean. **(2)**

.....

(b) Find the median. **(2)**

.....

(c) Write down the mode. **(1)**

.....

(d) Work out the range. **(1)**

.....

(Total for Question 1 is 6 marks)

2 The table shows the number of goals scored by a hockey team in each of 20 matches.

Goals	**0**	**1**	**2**	**3**	**4**
Frequency	3	8	6	2	1

(a) Write down the mode. **(1)**

.....

(b) Work out the median number of goals. **(2)**

.....

(c) Work out the mean number of goals. **(3)**

.....

(Total for Question 2 is 6 marks)

3 108 students were asked how they travel to school. The pie chart shows the results. 36 students walk.

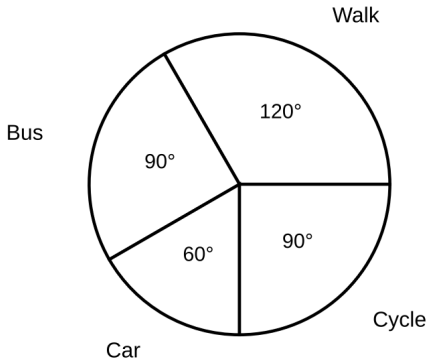


*Diagram **NOT** accurately drawn*

(a) Work out the number of students who travel by bus. **(2)**

.....

(b) What fraction of the students travel by car? Give your answer in its simplest form. **(2)**

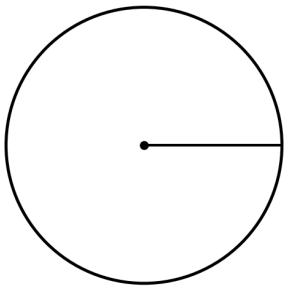
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(Total for Question 3 is 4 marks)

4 48 children were asked to choose their favourite fruit. The table shows the results.

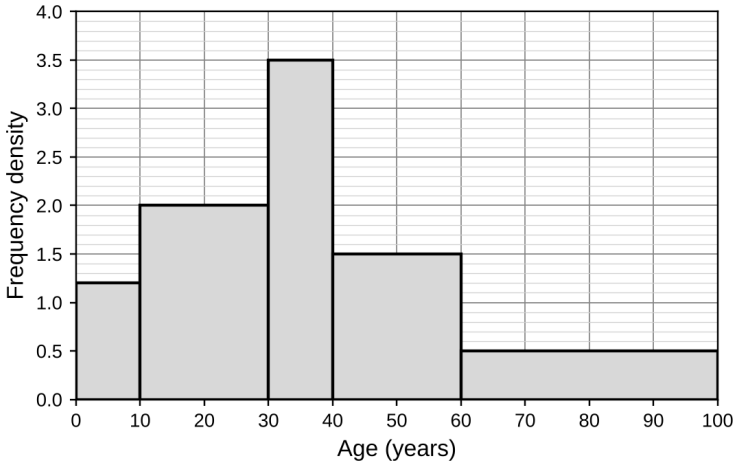
Fruit	**Apple**	**Banana**	**Orange**	**Cherry**
Number of children	18	12	6	12

Draw an accurate pie chart to show this information. **(4)**



(Total for Question 4 is 4 marks)

5 The histogram gives information about the ages of the people who visited a museum on Saturday.



(a) Work out the number of visitors aged more than 10 and up to 30. **(2)**

.....

(b) Work out an estimate for the number of visitors who were older than 50. **(3)**

.....

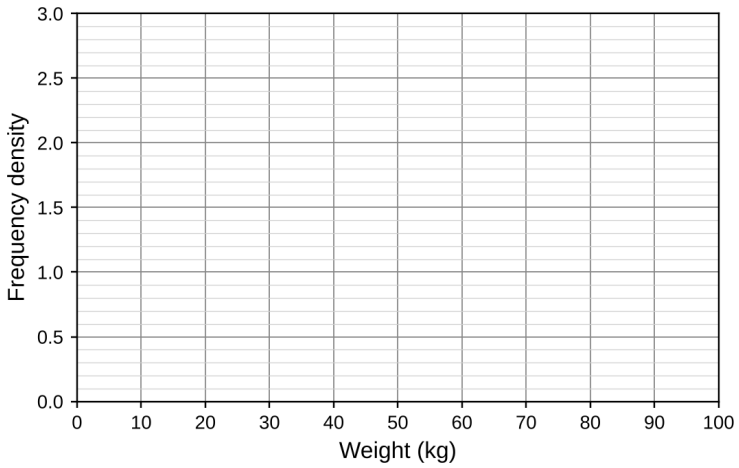
(Total for Question 5 is 5 marks)

6 The table shows information about the weights of 69 parcels.

(a) Complete the table by working out the frequency density for each class. **(2)**

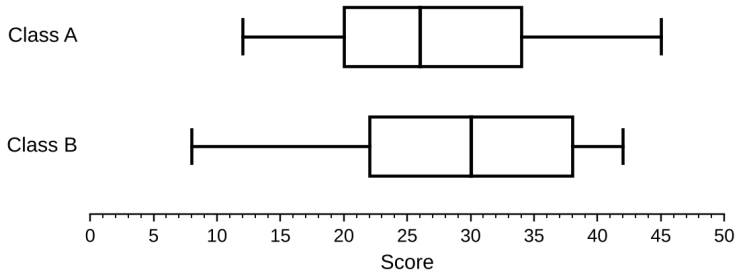
Weight (w kg)	**0 < w ≤ 20**	**20 < w ≤ 30**	**30 < w ≤ 40**	**40 < w ≤ 60**	**60 < w ≤ 100**
Frequency	6	12	25	18	8
Frequency density					

(b) Draw a histogram for this information. **(3)**



(Total for Question 6 is 5 marks)

7 The box plots show the marks scored by the students in Class A and in Class B in a test.



(a) Work out the interquartile range for Class A. **(2)**

.....

(b) Compare the distribution of the marks of Class A with the distribution of the marks of Class B. **(2)**

.....

.....

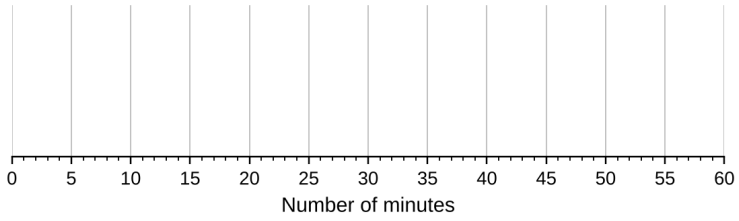
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(Total for Question 7 is 4 marks)

8 For a group of students, the lowest time to complete a puzzle was 5 minutes, the highest was 55 minutes, the median was 22 minutes, the lower quartile was 15 minutes and the upper quartile was 40 minutes.

On the grid, draw a box plot for this information. **(3)**



(Total for Question 8 is 3 marks)

9 A school has 1200 students. Zoe takes a stratified sample of 120 students by year group.

Complete the table to show the number of students in each year group that should be in the sample. **(3)**

Year group	**7**	**8**	**9**	**10**	**11**
Number of students	250	270	280	220	180
Number in sample					

(Total for Question 9 is 3 marks)

10 A lake contains N fish. 80 fish are caught, tagged and returned to the lake. A week later 60 fish are caught. 8 of these have tags.

(a) Work out an estimate for N. **(3)**

.....

(b) State one assumption you have made. **(1)**

.....

.....

(Total for Question 10 is 4 marks)

11 (a) The mean of five numbers is 8. A sixth number, 14, is added. Work out the mean of the six numbers. **(3)**

.....

(b) In Class A there are 20 students with a mean mark of 62. In Class B there are 30 students with a mean mark of 52. Work out the mean mark for all 50 students. **(3)**

.....

(Total for Question 11 is 6 marks)

12 (a) Dan asks, "Don't you agree that the new library is wonderful?" and gives the answers Yes / No. Write down one reason why this is a poor question. **(1)**

.....

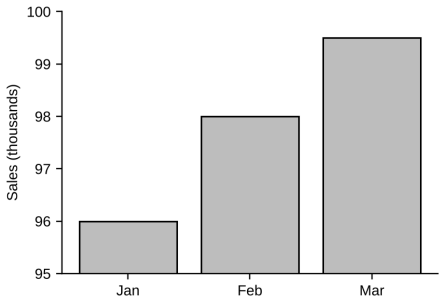
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(b) Alice wants to find out how much exercise students at her school do. She asks the first 20 students she sees outside the gym. Give one reason why her sample may not be representative. **(1)**

.....

.....

(c) The graph shows a company's sales. Explain why the graph could be misleading. **(2)**



.....

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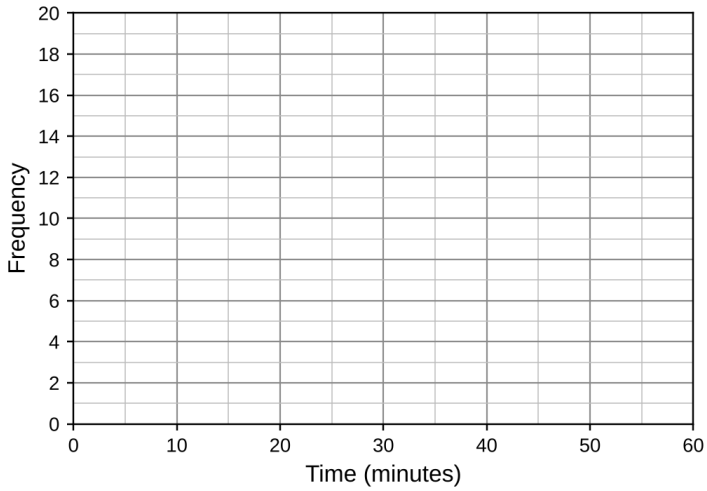
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(Total for Question 12 is 4 marks)

13 The table shows the times taken by 38 students to complete a puzzle.

Time (t minutes)	**0 < t ≤ 10**	**10 < t ≤ 20**	**20 < t ≤ 30**	**30 < t ≤ 40**	**40 < t ≤ 50**	**50 < t ≤ 60**
Frequency	3	7	12	9	5	2

On the grid, draw a frequency polygon for this information. **(2)**



(Total for Question 13 is 2 marks)

Section B: Calculator

A calculator may be used for questions 14 to 19.

14 The table shows the heights of 40 students.

Height (h cm)	**140 < h ≤ 150**	**150 < h ≤ 160**	**160 < h ≤ 170**	**170 < h ≤ 180**	**180 < h ≤ 190**
Frequency	4	10	16	8	2

(a) Write down the modal class. **(1)**

.....

(b) Write down the class interval that contains the median. **(1)**

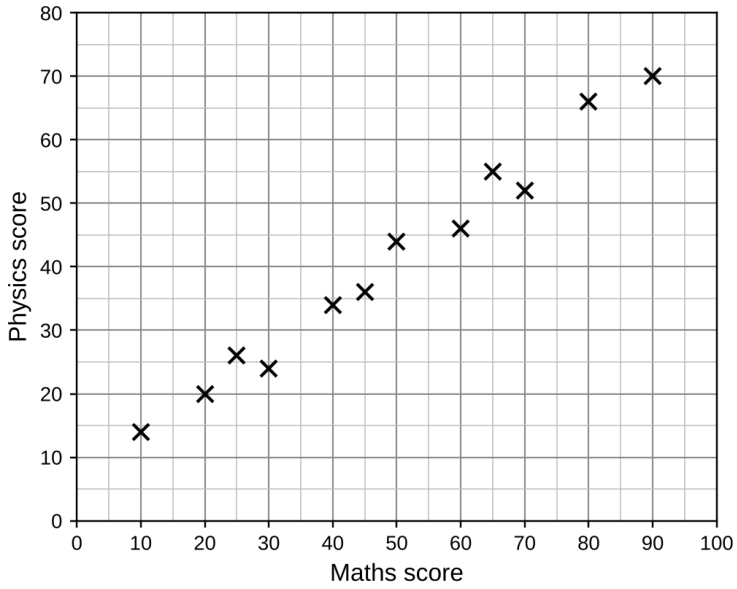
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(c) Work out an estimate for the mean height. **(4)**

.....

(Total for Question 14 is 6 marks)

15 The scatter graph shows the Maths score and the Physics score of 12 students.



(a) Write down the type of correlation. **(1)**

.....

(b) Draw a line of best fit on the scatter graph. **(2)**

(c) Use your line of best fit to estimate the Physics score of a student with a Maths score of 55. **(2)**

.....

(d) Explain why it would not be sensible to use your line to estimate the Physics score for a student with a Maths score of 120. **(1)**

.....

.....

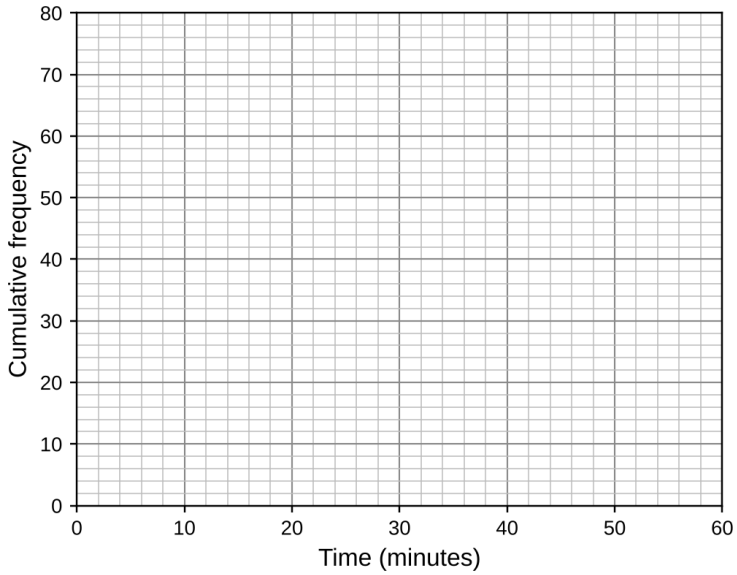
(Total for Question 15 is 6 marks)

16 80 runners took part in a race. The table shows information about their times.

(a) Complete the cumulative frequency table. **(2)**

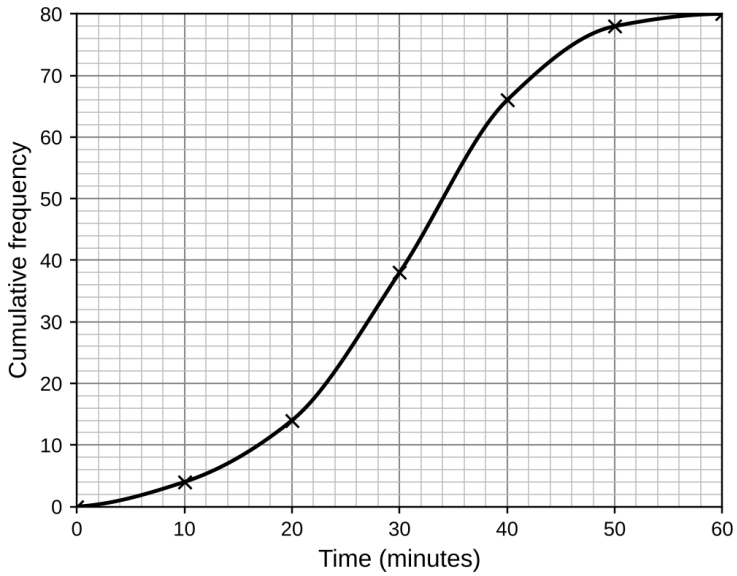
Time (t minutes)	**$0 < t \leq 10$**	**$10 < t \leq 20$**	**$20 < t \leq 30$**	**$30 < t \leq 40$**	**$40 < t \leq 50$**	**$50 < t \leq 60$**
Frequency	4	10	24	28	12	2
Cumulative frequency						

(b) On the grid, draw a cumulative frequency graph for this information. **(3)**



(Total for Question 16 is 5 marks)

17 The cumulative frequency graph shows the times taken by 80 runners in a different race.



(a) Use the graph to find an estimate for the median time. **(2)**

.....

(b) Use the graph to find an estimate for the interquartile range. **(3)**

.....

(c) Runners who take more than 45 minutes get a slow-runner badge. Use the graph to estimate how many runners get a badge. **(2)**

.....

(Total for Question 17 is 7 marks)

18 Team A scores a mean of 24 points per game with a range of 10 points. Team B scores a mean of 21 points per game with a range of 22 points.

Compare the performance of the two teams. **(2)**

.....

.....

.....

(Total for Question 18 is 2 marks)

19 The mean of 4 numbers is 12.5. The mean of 6 other numbers is 15.

Work out the mean of all 10 numbers. **(3)**

.....

(Total for Question 19 is 3 marks)

TOTAL FOR PAPER IS 85 MARKS

END

MARK SCHEME – Topic 6: Statistics

Guidance on marking: M marks are method marks, awarded for a correct method (even if the final answer is wrong). **A** marks are accuracy marks, awarded only if the relevant M mark has been gained. **B** marks are independent marks (no method needed). **C** marks are communication marks. **P** marks are process marks (a correct approach to a problem). **ft** = follow through from a previous incorrect answer. **oe** = or equivalent. **cao** = correct answer only. **SC** = special case. If the answer on the answer line differs from the working, the answer line is marked unless the working is clearly the intended answer. Correct answers with no working score full marks unless the question says 'You must show your working'.

Question	Working	Answer	Mark	Notes
1(a)	M1 $52 \div 8$ A1 6.5	6.5	2 AO1	
1(b)	M1 ordered list 3, 4, 5, 7, 7, 7, 9, 10 A1 7	7	2 AO1	
1(c)	B1 7	7	1 AO1	
1(d)	B1 $10 - 3 = 7$	7	1 AO1	
2(a)	B1 1	1	1 AO1	
2(b)	M1 10th and 11th values (cumulative frequencies 3, 11) A1 1	1	2 AO1	
2(c)	M1 $0 \times 3 + 1 \times 8 + 2 \times 6 + 3 \times 2 + 4 \times 1$ (use of fx) M1 $30 \div 20$ A1 1.5	1.5	3 AO1	
3(a)	M1 $36 \div 120 \times 90$ or $108 \div 360 \times 90$ A1 27	27	2 AO3	
3(b)	M1 $\frac{60}{360}$ A1 $\frac{1}{6}$	$\frac{1}{6}$	2 AO1	
4	B1 angle $= 360 \div 48 = 7.5^\circ$ per child B1 angles $135^\circ, 90^\circ, 45^\circ, 90^\circ$ M1 one sector drawn accurately (±2°) A1 all sectors and labels correct		4 AO1	
5(a)	M1 frequency = frequency density × class width: 2.0×20 A1 40	40	2 AO2	
5(b)	M1 $1.5 \times 10 = 15$ (50 to 60) M1 $0.5 \times 40 = 20$ (60 to 100) A1 35	35	3 AO3	
6(a)	B2 all correct: 0.3, 1.2, 2.5, 0.9, 0.2 (B1 for any 3 correct)		2 AO1	
6(b)	B1 bars correct widths (0–20, 20–30, 30–40, 40–60, 60–100) with no gaps B1 heights 0.3, 1.2, 2.5, 0.9, 0.2 (ft) B1 all five bars correct (±half a small square)		3 AO1	
7(a)	M1 $34 - 20$ A1 14	14	2 AO1	

Question	Working	Answer	Mark	Notes
7(b)	C1 median comparison: Class B's median (30) is higher than Class A's (26) C1 spread comparison: Class B's IQR (16) is larger than Class A's (14), so Class A's marks were slightly more consistent		2 AO3	
8	B1 correct scale and box from 15 to 40 B1 median line at 22 B1 whiskers to 5 and 55		3 AO1	
9	M1 $\frac{120}{1200} = \frac{1}{10}$ (sampling fraction) A1 25, 27, 28 A1 22, 18		3 AO3	
10(a)	M1 $\frac{8}{60} = \frac{80}{N}$ M1 $N = 80 \times 60 \div 8$ A1 600	600	3 AO3	
10(b)	C1 e.g. the tagged fish have mixed evenly with the others / no fish have died or been born or moved		1 AO3	
11(a)	M1 $5 \times 8 = 40$ M1 $(40 + 14) \div 6$ A1 9	9	3 AO3	
11(b)	M1 $20 \times 62 = 1240$ and $30 \times 52 = 1560$ M1 $(1240 + 1560) \div 50$ A1 56	56	3 AO3	
12(a)	C1 it is a leading question (suggests the answer) / biased		1 AO3	
12(b)	C1 students outside the gym are more likely to exercise (biased) / sample too small		1 AO3	
12(c)	C1 the vertical axis does not start at 0 (it starts at 95) C1 so the increase looks much bigger than it really is		2 AO3	
13	B1 plots at midpoints 5, 15, 25, 35, 45, 55 with correct heights 3, 7, 12, 9, 5, 2 (ft) B1 points joined with straight lines (not closed)		2 AO1	
14(a)	B1 $160 < h \leq 170$	160 < h ≤ 170	1 AO1	
14(b)	B1 $160 < h \leq 170$ (20th/21st values, cumulative frequencies 4, 14, 30)	160 < h ≤ 170	1 AO1	
14(c)	M1 midpoints 145, 155, 165, 175, 185 M1 $\sum fx = 580 + 1550 + 2640 + 1400 + 370 = 6540$ M1 $6540 \div 40$ A1 163.5	163.5 cm	4 AO1	
15(a)	B1 positive		1 AO1	
15(b)	M1 straight line drawn through (49, 41) (the mean point) or with 2 points above/below and a similar number of points on either side A1 line with a gradient of about 0.7 passing between (20, 20) and (80, 63)		2 AO2	

Question	Working	Answer	Mark	Notes
15(c)	M1 reading from 55 on the x-axis to the line A1 43 to 47		2 AO2	
15(d)	C1 120 is outside the range of the data (extrapolation)		1 AO3	
16(a)	B2 cumulative frequencies 4, 14, 38, 66, 78, 80 (B1 for 4 correct)		2 AO1	
16(b)	B1 plots at the *upper* class boundaries (10, 4), (20, 14) … (60, 80) (ft) B1 at least 5 points correct B1 smooth curve (or straight lines) through all points, starting at (0, 0)		3 AO1	
17(a)	M1 reads at 40 on the cumulative frequency axis A1 30 to 32		2 AO2	
17(b)	M1 reads at 20 and 60 M1 $Q_3 - Q_1$ with $Q_1 \approx 22$ and $Q_3 \approx 38$ A1 14 to 17		3 AO2	
17(c)	M1 reads at 45 minutes (about 72) A1 $80 - 72 = 8$ (accept 6 to 10)		2 AO3	
18	C1 Team A scores more on average (mean 24 > 21) C1 Team A is more consistent because its range is smaller (10 < 22)		2 AO3	
19	M1 $4 \times 12.5 = 50$ and $6 \times 15 = 90$ M1 $(50 + 90) \div 10$ A1 14	14	3 AO3	

Where marks are lost in Topic 6

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Averages from a frequency table:** mean = $\sum fx \div \sum f$. Median = the $\frac{n+1}{2}$th value. Do not add up the frequencies column and divide by the number of rows.

- ☐ **Grouped data:** use class mid-points to estimate the mean. The modal class is the one with the highest *frequency* (or highest frequency density in a histogram with unequal widths).

- ☐ **Histograms:** the *area* of the bar is the frequency. Frequency density = frequency ÷ class width.

- ☐ **Cumulative frequency:** plot against the *upper* class boundary. Median at $\frac{n}{2}$, quartiles at $\frac{n}{4}$ and $\frac{3n}{4}$.

- ☐ **Comparing data:** always compare an average (median/mean) *and* a measure of spread (IQR/range), and write your answer in the context of the question.

- ☐ **Pie charts:** angle = 360 ÷ total × frequency. Check the angles add to 360° before drawing.

- ☐ **Scatter graphs:** a line of best fit must be a single ruled straight line through the middle of the points. Do not extrapolate far beyond the data.

- ☐ **Sampling:** stratified sample = (group size ÷ population) × sample size. Capture–recapture: $N = \frac{\text{first catch} \times \text{second catch}}{\text{tagged in second catch}}$.